MATHEMATICAL OLYMPIAD PREPARATION

Four subject diagnostics

Release 0.1, print edition

Student instructions

1. Complete each subject as a separate sitting, preferably on different days.
2. Write a full mathematical argument for every claim. A correct answer without justification may earn only limited credit.
3. Use pen or pencil, plain paper and geometric drawing instruments where appropriate. Do not use a calculator, AI tool, internet source or hints during a diagnostic attempt.
4. If a solution is incomplete, record useful steps or observations. Stop the timed attempt when the allotted time ends.
5. Tell your mentor if you have seen a question or its underlying theorem before. You may reschedule or stop a session if needed.

Diagnostic schedule

Subject	Time	Marks
Algebra	45 minutes	14
Geometry	55 minutes	14
Number Theory	45 minutes	14
Combinatorics	45 minutes	14
Total	**190 minutes**	**56**

Notes

These papers contain eight independently attempted, proof-based practice questions. The scores are internal diagnostics. They are not official British Mathematical Olympiad or International Mathematical Olympiad results, and cannot be converted into a selection ranking.

Answers can be written on separate sheets or in the optional answer booklet. The complete examiner's solutions are provided only to the mentor.

Independent formative material, 19 October 2026. No official UKMT or IMO affiliation.

Algebra: Diagnostic A

Time allowed: 45 minutes **Total:** 14 diagnostic marks

Instructions Answer both questions. Show full working and give complete proofs. No calculator, internet, AI assistance or hints during an independent diagnostic. Use separate answer paper if required.

Candidate: _____ Date: _____

Question 1 [7 marks]

ALG-01

Let $f : \mathbb{Z} \to \mathbb{Z}$ satisfy

$$f(x + y) + f(x - y) = 2f(x) + 2f(y)$$

for every pair of integers x and y.

Determine every such function f, and prove that your list is complete.

Question 2 [7 marks]

ALG-02

Let a, b, c be positive real numbers. Prove that

$$(a + b + c)^3 + 9abc \geq 4(a + b + c)(ab + bc + ca).$$

Determine all cases of equality.

End of paper. Write your answers on separate paper, labelled with the question number.

Geometry: Diagnostic G

Time allowed: 55 minutes **Total:** 14 diagnostic marks

Instructions Answer both questions. Show full working and give complete proofs. No calculator, internet, AI assistance or hints during an independent diagnostic. Use separate answer paper if required.

Candidate: _____ Date: _____

Question 1 [7 marks]

GEO-01

In an acute triangle ABC, let E be the foot of the perpendicular from B to AC, and let F be the foot of the perpendicular from C to AB. Let ℓ be the tangent at A to the circumcircle of ABC.

Prove that EF is parallel to ℓ. You may use a clearly labelled diagram.

Question 2 [7 marks]

GEO-02

Let ABC be a nondegenerate triangle, and let P be a point on its circumcircle, distinct from A, B, and C. Let X, Y, Z be the perpendicular projections of P onto the *lines* AB, BC, and CA, respectively. The feet may lie on extensions of the sides.

Prove that X, Y, Z are collinear. Address any coincident-foot cases.

End of paper. Write your answers on separate paper, labelled with the question number.

Number Theory: Diagnostic N

Time allowed: 45 minutes **Total:** 14 diagnostic marks

Instructions Answer both questions. Show full working and give complete proofs. No calculator, internet, AI assistance or hints during an independent diagnostic. Use separate answer paper if required.

Candidate: _____ Date: _____

Question 1 [7 marks]

NT-01

Determine all integer triples (x, y, z) satisfying

$$x^2 + y^2 = 3z^2.$$

Prove that no other integer solutions exist.

Question 2 [7 marks]

NT-02

For an integer $N \neq 0$, let $v_3(N)$ denote the largest nonnegative integer e for which 3^e divides N.

For every integer $k \geq 0$, determine

$$v_3\left(2^{3^k} + 1\right).$$

Prove your formula without quoting the lifting-the-exponent lemma.

End of paper. Write your answers on separate paper, labelled with the question number.

Combinatorics: Diagnostic C

Time allowed: 45 minutes **Total:** 14 diagnostic marks

Instructions Answer both questions. Show full working and give complete proofs. No calculator, internet, AI assistance or hints during an independent diagnostic. Use separate answer paper if required.

Candidate: _____ Date: _____

Question 1 [7 marks]

COMB-01

Write the integers $1, 2, 3, \ldots, 2026$ in a row. In one move, choose any two adjacent entries a and b and replace them with the single entry

$$a + b + ab.$$

Continue until only one entry remains.

Determine the final entry, and prove that it does not depend on the choices of adjacent pairs.

Question 2 [7 marks]

COMB-02

In a group, every pair of distinct people are either mutual friends or mutual strangers. Find the *smallest* positive integer n with the following property:

> *Every group of n people contains either three pairwise mutual friends or three pairwise mutual strangers.*

Prove that your value works, and that a smaller value does not.

End of paper. Write your answers on separate paper, labelled with the question number.